Supplementary Materials for
Navigating high-dimensional spaces in chemistry and beyond with a reference-based coordinate assignment

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Computational Details

The hyperparameter tuning and training of the machine learning models were conducted in the JUSTUS2 cluster. The results were postprocessed on a DELL XPS 15 9510 laptop with an 11th Gen Intel(R)Core(TM) i7-11800H @ 2.30 GHz, 64-bits operating system (x64-based processor), and 32.0 GB of installed RAM.

A. Molecular Representations

The SOAP representation (*1*) was generated using the DScribe package (*2*) with a cutoff radius of 6 Å, using a set of 8 radial basis functions with Gaussian-type orbitals, and a maximum degree of spherical harmonics set to 100. The Coulomb Matrix (*3*) was generated using the DScribe package and using the L2 norm for sorting. The global SLATM (*4*) representation was generated using the QML code (*5*) with parameters sigmas=[0.05,0.05], dgrids=[0.03,0.03], rcut=4.8, and rpower=6. The MAOC and PC3-MAOC representations (*6*) were generated via the maoc-support-functions package, using the pcseg-0 basis set.

B. Dimensionality Reduction Techniques

Four dimensionality reduction techniques were used in this study. The scikit-learn library (*7*) was used to perform MDS, with n components set to 20 and Euclidean dissimilarity. Scikit-learn was also used to perform PCA, with the number of dimensions changing based on the specific application. The MDS with reference points code was modified from the original version for a more direct comparison with the RCA method. The points that were closest to the centroids of 20 clusters identified using KMeans (*8*) were chosen as n reference points. Then, distances were calculated between reference points and between reference points and non-reference points. The non-anchor points were projected into the reduced space by solving a least-squares system, and the reference points were embedded using classical MDS. Scikit-learn was used to compute the pairwise distance matrices and for the KMeans clustering. The RCA code was solely written using the NumPy package (*9*) and generated using the RCA Python package. This package contains a single function, RCA vectorised, which requires the reference constellation (ref const) in its reduced form and the distances between the reference points and the unknown points (distances).

C. Machine Learning

Five machine learning models were employed to evaluate the performance of RCA spaces across regression and classification tasks. For regression, kernel ridge regression (KRR) with a Laplacian kernel was used. The kernel width σ was optimised over the range 1–10 000 in increments of 1 000, and the regularisation parameter λ was selected from $\{1.0 \times 10^{-10}, 3.2 \times 10^{-8}, 1.0 \times 10^{-5}, 3.2 \times 10^{-3}, 1.0\}$. Hyperparameter optimisation was performed on 30% of the dataset, and final model training used an 80/20 train–test split. All KRR calculations were conducted using the QML package (*5*) without modification.

To assess the applicability of RCA beyond chemical datasets, image and audio classification benchmarks were considered. For image classification, two models were employed: a Nu-support vector classifier (Nu-SVC) and a feed-forward neural network. The Nu-SVC implementation from Scikit-learn was used without modification.

A multimodal learning framework was developed to evaluate whether combining original image features with RCA representations improves predictive accuracy (fig. S1). One neural network processed the full image inputs, while a second network operated on the RCA-reduced representations. Their outputs were concatenated and passed to a third feed-forward neural network for final classification. Each subnetwork consisted of three hidden layers. For the full-image network, hidden layer sizes of 300, 400, and 500 neurons were evaluated. For the RCA-based network, hidden layer sizes between 20 and 200 neurons were explored. The fusion network employed identical layer sizes selected from $\{50, 80, 100\}$. Both Adam and stochastic gradient descent (SGD) optimisers were assessed. The reference constellation for images was constructed using 20 centroids, followed by PCA-based dimensionality reduction.

Random Forest hyperparameters were optimised over the following ranges: number of estimators $\in \{100, 200, 300, 400, 500\}$, maximum depth $\in \{\texttt{None}, 10, 20, 30, 40\}$, minimum samples split $\in \{2, 5, 10\}$, minimum samples leaf $\in \{1, 2, 4\}$, maximum features $\in \{\texttt{sqrt}, \texttt{log2}, \texttt{None}\}$, and bootstrap $\in \{\texttt{True}, \texttt{False}\}$.

A multimodal audio classification scheme was further implemented by training separate Random Forest classifiers on MFCC features and RCA spaces. Their predicted class probabilities were concatenated to form meta-features, which served as input to a multinomial logistic regression

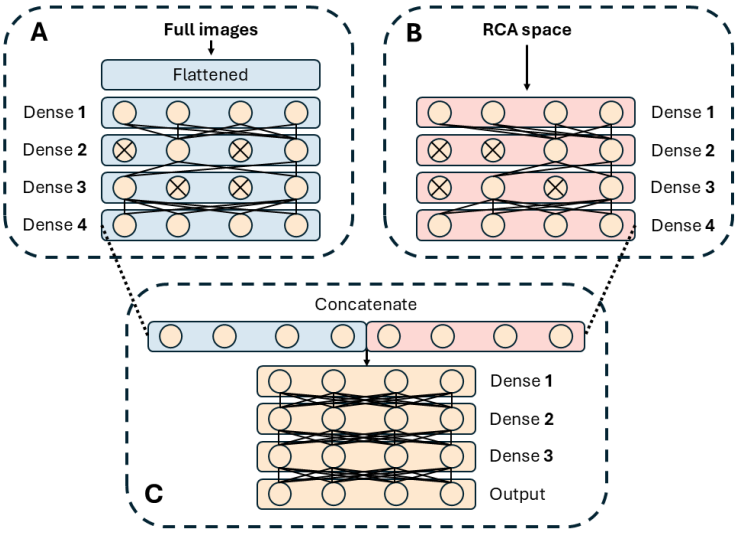

Figure S1: **Structure of the multimodal learning architecture used for classification of the Fashion-MNIST dataset images.**

classifier. The logistic regression hyperparameters were optimised over $C \in logspace(-3, 3, 10)$ with solvers $\in \{\texttt{lbfgs}, \texttt{saga}\}$. All classification models were implemented using Scikit-learn without modification.

D. Datasets

The applicability of RCA representations in chemical machine learning is evaluated across five datasets. Among these, the N-HPC-1X dataset is introduced here for the first time. The N-HPC-1X dataset is generated by doping frameworks containing four to eight conjugated rings with one to six nitrogen atoms using the GANNA algorithm (*6*). This dataset contains 10,582 nitrogen-doped compounds in their neutral closed-shell configuration. The geometry of the compounds, along with the single point energy and the energies of HOMO and LUMO, was computed using the PBE0-D3/def2-TZVP level of the theory. The compositional distribution of the framework dataset is shown in fig. S2, and the corresponding ranges of computed properties are presented in fig. S3.

The COMPAS-1D dataset (*10*) comprises 8,678 polycyclic aromatic hydrocarbons with optimized geometries and single-point energies computed at the B3LYP-D3(BJ)/def2-TZVP level of theory. Among the available properties, only the HOMO energy, LUMO energy, HOMO–LUMO gap, and the single-point energy of the neutral species were considered in this study.

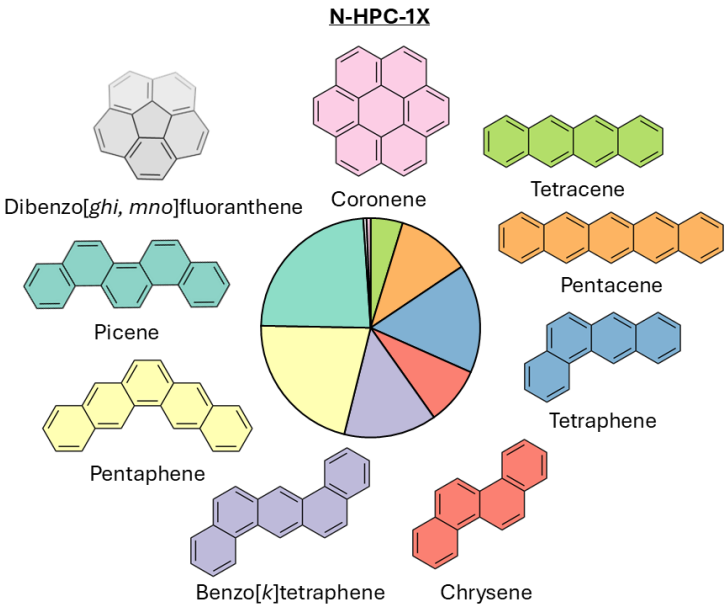

Figure S2: **The framework distribution of the N-HPC-1X dataset.**

The PEPCONF dataset (*11*) comprises 3,775 peptide structures, including dipeptides, tripeptides, disulfide-bridged peptides, bioactive peptides, and cyclic peptides. In this work, the target property is the reference energy computed at the LC-ωPBE-XDM/aug-cc-pVTZ level of theory.

The transition metal quantum mechanics (tmQM) dataset (*12*) comprises 86,665 mononuclear transition metal complexes. For each complex, the HOMO energy, LUMO energy, and HOMO–LUMO gap were computed at the TPSSh-D3(BJ)/def2-SVP level of theory, while the polarisability was obtained using GFN2-xTB (*13*). In this study, a subset of 10,543 complexes was considered, for which calculations were performed using the pcseg-0 basis set.

A subset of 6,095 constitutional isomers of $C_7H_{10}O_2$ from the QM9 dataset (*14*) was used in this study. The HOMO energy, LUMO energy, and HOMO–LUMO gap were computed at the B3LYP/6-31G(2df,p) level of theory, while single-point energies were obtained at the G4MP2 level.

To evaluate the generality of RCA beyond chemical applications, two non-chemical benchmark datasets were examined. The Fashion-MNIST dataset (*15*) comprises 70,000 grayscale images of size 28 × 28 pixels, distributed across 10 classes: *T-shirt/top*, *Trouser*, *Pullover*, *Dress*, *Coat*, *Sandal*, *Shirt*, *Sneaker*, *Bag*, and *Ankle boot*.

The Google Speech Commands dataset (*16*) consists of one-second WAV recordings of spoken English commands. From the full dataset, 38,546 samples were selected, each assigned to one of

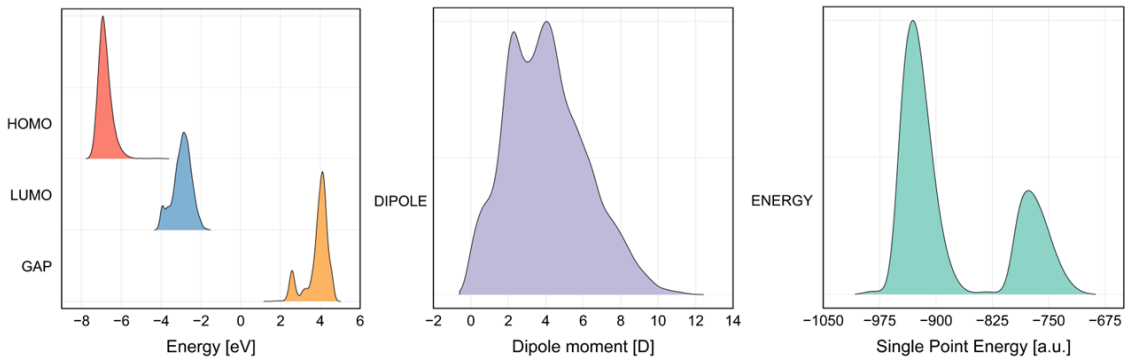

Figure S3: **The distribution of property values for the HOMO energy, LUMO energy, HOMO-LUMO gap (left), dipole moment (middle), and single point energy (right) of the N-HPC-1X dataset.**

10 classes: *yes*, *no*, *up*, *down*, *left*, *right*, *on*, *off*, *stop*, and *go*. Audio signals were processed using the Librosa library: recordings were normalised and transformed into 20 Mel-frequency cepstral coefficients (MFCCs), which were subsequently flattened to form feature vectors.

Supplementary Text

S1. Mathematical Formulation of RCA

RCA constructs the projected space through the following steps:

1. Select the subset of points that will serve as the reference constellation.

2. Calculates the distance between each point in the X subset and each point in the reference constellation using their original N-dimensional coordinates.

3. Apply a dimensionality reduction technique to obtain a lower-dimensional representation of the coordinates of the points in the reference constellation, with the dimensionality of the embedding set equal to the number of compounds in the reference constellation.

4. Using the reference coordinates calculated in Step 3 and the distances from Step 2, compute the distance equations between the new reference constellation coordinates and the unknown

coordinates of the points in the X subset:

$$\sum_{\alpha=1}^{n(Y)} (x_\alpha - X_{i\alpha})^2 = r_i^2 \quad \text{for } i = 1, 2, \ldots, n(Y) \tag{S1}$$

Each point in the X subset is represented by coordinates $(x_1, x_2, \ldots, x_{n(Y)})$, and each point α in the reference constellation by $(X_{i\alpha}, X_{2\alpha}, \ldots, X_{n(Y)\alpha})$;

5. Subtract the distance equation of the first reference point from each of the others to obtain a system of $N - 1$ linear equations of the form:

$$\sum_{\alpha=1}^{n(Y)} (x_{i\alpha} - X_{1\alpha}) x_\alpha = \frac{1}{2} \left(\sum_{\alpha=1}^{n(Y)} (x_{i\alpha}^2 - X_{1\alpha}^2) - (r_i^2 - r_1^2) \right) \tag{S2}$$

6. Solve the system from Step 5 by expressing $N - 1$ coordinates as linear functions of the last coordinate. For example, the equation for the $(N - 1)$-th coordinate can be written as:

$$x_{N-1} = X_{N-1} x_N + \beta_{N-1} \tag{S3}$$

Find x_N by substituting x_α with $X_\alpha x_N + \beta_\alpha$ into the equation from Step 4, which transforms it into a quadratic form. Under the tangency assumption, this quadratic equation has a single solution (its discriminant equals zero);

7. Substitute the value x_N from Step 6 into the linear equations to compute all remaining coordinates.

S2. Comparison Between RCA, Anchor-Based MDS, and ChemGPS

The use of reference points and multilateration for dimensionality reduction is not unique to RCA and has appeared in both general data analysis and chemical informatics. Landmark-based multidimensional scaling (MDS with reference points) (*17, 18*) selects a subset of points as fixed landmarks, performs classical MDS on this subset, and subsequently positions the remaining points by triangulation using their distances to the reference points. Both RCA and anchor-based MDS therefore construct reduced representations relative to a predefined subset of reference points.

Although RCA and Landmark MDS share procedural similarities, particularly the use of a subset of reference points and the linear transformation of distances, their conceptual foundations differ.

LMDS is a spectral approximation of classical multidimensional scaling, whereas RCA constructs a reference-defined coordinate system without attempting to approximate a global eigenstructure.

In chemical space analysis, ChemGPS (*19*) was introduced to facilitate navigation and extrapolation. ChemGPS employs predefined "satellite" compounds, chosen to span extreme regions of property space, together with core compounds, and applies principal component analysis (PCA) to construct a chemically interpretable coordinate system. Although ChemGPS shares the use of reference structures and PCA-based compression, its primary objective is the management of chemographic topology and extrapolation. RCA, by contrast, reformulates dimensionality reduction itself as a positioning problem, assigning coordinates directly through distances to reference points rather than through variance maximisation or projection.

S3. Comparative Analysis of PCA, MDS, and Anchor-Based MDS with 20 reference points

Principal component analysis (*20, 21*) (PCA) and multidimensional scaling (*22*) (MDS) are among the most widely used dimensionality reduction techniques. In this section, we present a comparative analysis of PCA, MDS, anchor-based MDS (with 20 reference points), and RCA. The techniques are evaluated with respect to their ability to preserve pairwise distances in high-dimensional space, as well as their impact on predictive performance when used as inputs to machine learning models.

Visual comparison of reduced spaces using two-dimensional projections provides limited insight due to the substantial loss of structural information inherent in such representations. A more informative criterion for assessing dimensionality reduction techniques is their capacity to preserve the original pairwise distances between data points. To quantify this, the distance matrix of the reduced space is compared directly with the distance matrix of the original high-dimensional space by analysing their differences. figure S4 presents the flattened upper-triangular elements of the difference matrices obtained by comparing the full SOAP representation with the corresponding reduced-space representations for the N-HPC-1X dataset.

When reduced representations are used as inputs to machine learning models, predictive accuracy provides a practical measure of the quality of the embedding, provided that performance remains comparable to that achieved with the full, unreduced representation.

Table S1 reports the kernel ridge regression (KRR) accuracy for predicting the HOMO–LUMO gap of the N-HPC-1X dataset using PCA, MDS, RCA with 20 reference points, and anchor-based

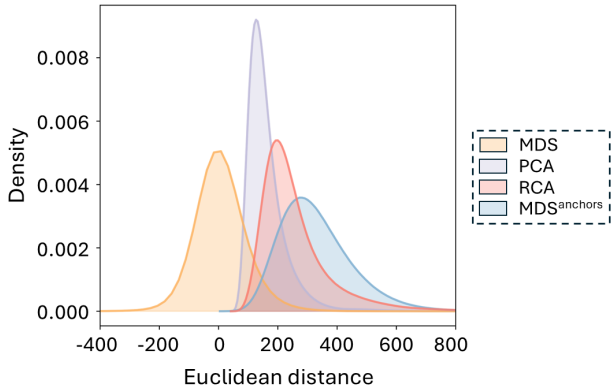

Figure S4: **The differences in pairwise distance distributions of compounds in the N-HPC-1X dataset between the original high-dimensional SOAP space and the spaces obtained after applying dimensionality reduction.**

Table S1: **The mean absolute error in predicting the HOMO-LUMO gap of the compounds in the N-HPC-1X dataset and the output generation time of the reduced space when using the four selected dimensionality reduction techniques.**

Method	MAE [eV]	Time of generation [s]
PCA	0.082	73.00
MDS	0.138	3180.00
MDS with 20 reference points	0.088	940.00
RCA	0.083	0.25

MDS with 20 reference points. Relative to the full SOAP representation (MAE = 0.054 eV), PCA, RCA, and anchor-based MDS yield comparable predictive performance, whereas standard MDS exhibits a modest reduction in accuracy.

S4. Impact of the Number of Cluster Centroids in the Reference Constellation

When cluster centroids are used as the reference constellation, the choice of their number has a significant impact on the predictive accuracy of the machine learning model. Figure S5 shows the decrease in the KRR MAE for predicting the HOMO–LUMO gap of compounds in the N-HPC-1X

dataset using the SOAP representation.

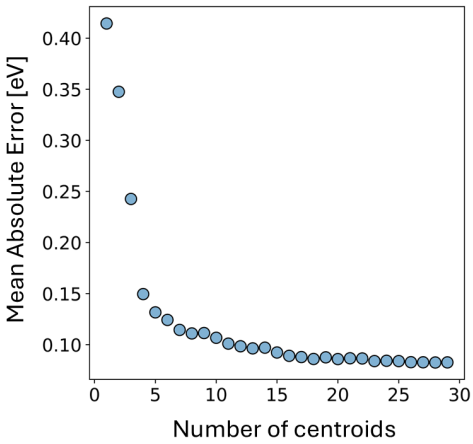

Figure S5: **Test set MAE for HOMO–LUMO gap prediction on the N-HPC-1X dataset as a function of the number of cluster centroids using the KRR model.** The RCA used as input for the KRR model is PCA-based, and the reported MAE values correspond to the test set from an 80:20 data split using 5-fold cross validation.

Unless otherwise specified, the number of cluster centroids used in this work is 20.

S5. Machine Learning Performance Using the Full Representation Spaces

To enable a direct comparison with the RCA spaces, this section reports the predictive accuracy obtained when the full representation spaces are used as input to the machine-learning models for the N-HPC-1X, COMPAS-1D, PEPCONF, tmQM, QM9, Fashion-MNIST, and Google Speech Commands datasets.

S6. Machine Learning Performance Using the RCA Spaces

In the main text, only a limited set of benchmarks is presented to illustrate the performance of RCA spaces in property prediction. In this section, we provide a comprehensive evaluation of RCA spaces used as input for machine-learning models across the N-HPC-1X, COMPAS-1D, PEPCONF, tmQM, and QM9 datasets.

Table S2: **Mean absolute error of the KRR models reported for inputs consisting of the full feature spaces derived from the specified molecular representations for the N-HPC-1X dataset.**

Input	HOMO [eV]	LUMO [eV]	GAP [eV]	SPE [a.u.]
CM	0.138	0.121	0.209	3.361
PC3-MAOC	0.123	0.134	0.186	0.228
MAOC	0.120	0.132	0.174	0.125
SOAP	0.047	0.025	0.054	0.166
SLATM	0.048	0.032	0.059	0.021

Table S3: **Mean absolute error of the KRR models reported for inputs consisting of the full feature spaces derived from the specified molecular representations for the COMPAS-1D dataset.**

Input	HOMO [eV]	LUMO [eV]	GAP [eV]	SPE [a.u.]
CM	0.098	0.101	0.195	104.857
PC3-MAOC	0.068	0.072	0.136	7.685
MAOC	0.060	0.067	0.123	4.519
SOAP	0.033	0.033	0.064	54.026
SLATM	0.033	0.037	0.066	0.057

Table S4: **Mean absolute error of the KRR models reported for inputs consisting of the full feature spaces derived from the specified molecular representations for the 6069 constitutional isomers of QM9 dataset.**

Input	HOMO [eV]	LUMO [eV]	GAP [eV]	SPE [a.u.]
CM	0.006	0.012	0.014	0.017
PC3-MAOC	0.006	0.014	0.015	0.029
MAOC	0.008	0.030	0.030	0.024
SOAP	0.005	0.008	0.009	0.029
SLATM	0.003	0.004	0.004	0.002

Table S5: **Mean absolute error of the KRR models reported for inputs consisting of the full feature spaces derived from the specified molecular representations for the conformational energies of the PEPCONF dataset.**

Input	$E_{conf}[eV]$
CM	0.124
PC3-MAOC	0.138
MAOC	0.122
SOAP	0.114
SLATM	0.082

Table S6: **Mean absolute error of the KRR models reported for inputs consisting of the full PC3-MAOC-derived feature spaces of the tmQM dataset.**

Input	HOMO [eV]	LUMO [eV]	GAP [eV]	Polarisability $[Cm^2V^{-1}]$
PC3-MAOC	0.043	0.043	0.028	12.255

Table S7: **Difference between the results obtained using full images as input and RCA-reduced arrays.** A Nu-SVC model with an RBF kernel was employed, following optimisation of the kernel space for classification of the Fashion-MNIST labels.

	t-shirt	**trouser**	**pullover**	**dress**	**coat**	**sandal**	**shirt**	**sneaker**	**bag**	**ankle boot**
t-shirt	**-38**	0	-16	29	-1	31	-36	0	31	0
trouser	1	**59**	-15	-48	0	5	-10	0	8	0
pullover	5	0	**-34**	1	18	32	-54	0	32	0
dress	-6	2	0	**3**	-7	20	-12	0	7	0
coat	3	0	-56	31	**55**	15	-55	0	7	0
sandal	1	0	-2	-1	0	**-65**	-1	12	44	12
shirt	-18	0	-26	16	-25	32	**-23**	0	44	0
sneaker	0	0	0	0	0	-14	0	**36**	2	-24
bag	-3	0	-22	11	1	20	-29	9	**13**	0
ankle boot	0	0	1	2	0	16	7	19	77	**-122**

Table S8: **Difference between the results obtained using full images as input and the multi-modal learning approach.** A neural network model was employed following optimisation of the architecture and hyperparameters for classification of the Fashion-MNIST labels.

	t-shirt	**trouser**	**pullover**	**dress**	**coat**	**sandal**	**shirt**	**sneaker**	**bag**	**ankle boot**
t-shirt	**-1615**	9	-25	-208	-3	2	-529	0	15	-1
trouser	25	**-155**	0	-34	-5	0	21	0	-1	10
pullover	59	3	**-66**	-71	49	0	467	0	29	3
dress	299	122	-9	**460**	-81	0	322	0	56	52
coat	200	5	49	-25	**-90**	0	579	0	42	0
sandal	1	-2	0	-7	0	**74**	-3	4	100	57
shirt	1011	1	65	-61	138	2	**-845**	0	36	1
sneaker	0	0	0	-3	0	-63	0	**49**	-9	105
bag	11	-3	-15	-38	-7	0	-11	4	**-291**	7
ankle boot	9	0	0	-12	0	-16	0	-56	20	**-234**

Table S9: **Difference between the results obtained using pre-processed audio signals as input and RCA-reduced arrays for the Google Speech Commands dataset (RF model).**

	Yes	**No**	**Up**	**Down**	**Left**	**Right**	**On**	**Off**	**Stop**	**Go**
Yes	**42**	-4	0	-2	-27	-6	0	2	-3	-2
No	1	**31**	-11	12	-2	-4	-9	-3	2	-17
Up	-4	-6	**-20**	14	25	-12	-6	3	13	-7
Down	2	-21	10	**42**	2	0	-9	-1	-4	-21
Left	-30	-21	-6	2	**110**	-42	-1	-13	-4	5
Right	-16	-28	-2	0	-38	**155**	-27	-34	-7	-3
On	-5	-18	1	-13	11	4	**71**	-11	3	-43
Off	-6	-5	-6	0	-4	-12	-28	**57**	9	-5
Stop	0	4	0	16	11	-2	-5	-9	**-12**	-3
Go	0	-75	-2	-54	9	-5	-25	2	-1	**151**

Table S10: **Difference between the results obtained from the use of pre-processed audio signals as input and the stacked random forests approach.**

	Yes	**No**	**Up**	**Down**	**Left**	**Right**	**On**	**Off**	**Stop**	**Go**
Yes	**8**	-4	6	-1	-10	3	0	5	-5	-2
No	-1	**-44**	2	28	-4	2	3	0	4	10
Up	-2	-8	**-47**	5	16	-6	11	19	18	-6
Down	7	-20	17	**8**	-8	4	4	5	0	-17
Left	-5	-18	10	3	**-3**	-4	-5	12	3	7
Right	1	-11	5	12	4	**-11**	0	-2	1	1
On	3	-10	7	0	12	4	**-21**	-1	6	0
Off	1	1	9	1	12	1	1	**-37**	8	3
Stop	-3	10	-3	10	13	-3	-2	4	**-26**	0
Go	1	-46	-8	8	8	-1	-3	0	0	**41**

Table S11: **Mean absolute error of the KRR models reported for inputs consisting of the RCA spaces derived from the specified molecular representations for the N-HPC-1X dataset.** Green indicates an increase in accuracy compared to the full spaces, yellow indicates a decrease in accuracy of 0.1 eV/a.u. or less, and red indicates a decrease in accuracy of more than 0.1 eV/a.u.

Input	HOMO [eV]	LUMO [eV]	GAP [eV]	SPE [a.u.]
CM	0.157	0.169	0.258	1.986
PC3-MAOC	0.141	0.158	0.226	0.071
MAOC	0.150	0.177	0.258	0.048
SOAP	0.077	0.072	0.108	0.714
SLATM	0.092	0.086	0.115	0.043

Table S12: **Mean absolute error of the KRR models reported for inputs consisting of the RCA spaces derived from the specified molecular representations for the COMPAS-1D dataset.** Green indicates an increase in accuracy compared to the full spaces, yellow indicates a decrease in accuracy of 0.1 eV/a.u. or less, and red indicates a decrease in accuracy of more than 0.1 eV/a.u.

Input	HOMO [eV]	LUMO [eV]	GAP [eV]	SPE [a.u.]
CM	0.103	0.111	0.277	11.536
PC3-MAOC	0.084	0.093	0.175	1.449
MAOC	0.085	0.092	0.174	0.764
SOAP	0.066	0.069	0.134	141.663
SLATM	0.049	0.051	0.098	0.142

Table S13: **Mean absolute error of the KRR models reported for inputs consisting of the RCA spaces derived from the specified molecular representations for the 6069 constitutional isomers of QM9 dataset.** Green indicates an increase in accuracy compared to the full spaces, yellow indicates a decrease in accuracy of 0.1 eV/a.u. or less.

Input	HOMO [eV]	LUMO [eV]	GAP [eV]	SPE [a.u.]
CM	0.009	0.032	0.032	0.032
PC3-MAOC	0.008	0.030	0.031	0.025
MAOC	0.008	0.032	0.033	0.025
SOAP	0.007	0.015	0.018	0.019
SLATM	0.005	0.006	0.008	0.005

Table S14: **Mean absolute error of the KRR models reported for inputs consisting of the RCA spaces derived from the specified molecular representations for the conformational energies of the PEPCONF dataset.** Green indicates an increase in accuracy compared to the full spaces, yellow indicates a decrease in accuracy of 0.1 eV/a.u. or less.

Input	$E_{conf}[eV]$
CM	0.127
PC3-MAOC	0.121
MAOC	0.135
SOAP	0.133
SLATM	0.104

Table S15: **Mean absolute error of the KRR models reported for inputs consisting of the RCA spaces of the tmQM dataset.** Green indicates an increase in accuracy compared to the full spaces, yellow indicates a decrease in accuracy of 0.1 eV/a.u. or less, and red indicates a decrease in accuracy of more than 0.1 eV/a.u.

Input	HOMO [eV]	LUMO [eV]	GAP [eV]	Polarisability [Cm^2V^{-1}]
PC3-MAOC	0.042	0.042	0.029	14.329

S7. Introducing Chemically Informative Structures for Constructing the Reference Constellation

The selection of the reference constellation is user-defined, subject to the equidistance criterion outlined in the main text. While domains such as images or audio may not possess inherently scientifically meaningful fragments, the macromolecules in the N-HPC-1X and COMPAS-1D datasets are systematically constructed from mono-, di-, and tricyclic building blocks. By incorporating such chemically interpretable fragments into the reference constellation, RCA can be reformulated as a fragmentation-based framework. Figure S6 presents a schematic representation of the transformation in the geometry of chemical space as the nature of the reference compounds is varied.

The N-HPC-1X dataset comprises nitrogen-doped polycyclic organic skeletons. To construct a chemically informative reference set, the simplest mono-, di-, and tricyclic scaffolds were generated and subsequently doped using the GANNA algorithm (*6*). This procedure yielded 9 N-doped monocycles, 50 bicycles, and 842 tricycles. In contrast, the COMPAS-1D dataset contains exclusively all-carbon analogues, resulting in only four corresponding fragments: one monocycle (benzene), one bicycle (naphthalene), and two tricycles (anthracene and phenanthrene).

For the reduction of the chemical space of the N-HPC-1X dataset, each fragment class was placed separately into its own reference constellation. In contrast, for the COMPAS-1D dataset, the limited number of reference compounds allowed all fragments to be combined into a single reference constellation.

In the case of the N-HPC-1X dataset, expansion of the reference constellation leads to increased structural similarity among the fragments, resulting in progressively smaller distances between the

reference points and previously unseen compounds.

Table S16 reports the KRR performance obtained for RCA spaces constructed from different underlying representations and evaluated across multiple properties. In general, the use of RCA spaces as inputs to machine-learning models results in a modest decrease in predictive accuracy relative to the full representation space. The most pronounced loss in accuracy is observed for single-point energies. In contrast, for orbital energies and related energy differences, the reduction in accuracy is in some cases below 0.1 eV. Notably, for SLATM-based models targeting properties such as the LUMO energy and the HOMO–LUMO gap, RCA spaces can even surpass the predictive performance of the full SLATM representation.

The reduced predictive performance of the RCA spaces across representations can be attributed to the positioning of the reference constellation relative to the unknown compounds in the original high-dimensional space. Consequently, the RCA space fails to faithfully preserve the structural relationships present in the original representation, resulting in diminished machine-learning accuracy.

Support for this interpretation is provided by the systematic improvement observed for certain representations and target properties as the fragment size increases. In particular, constructing the reference constellation from tricyclic fragments mitigates equidistance violations, as these motifs more closely approximate the structural complexity of compounds in the N-HPC-1X dataset.

The equidistance limitation becomes even more pronounced for the COMPAS-1D dataset, which comprises large PAH structures positioned substantially far from the chosen reference constellation. Under these conditions, the geometric distortion introduced in the RCA space is amplified. Consequently, when these spaces are used as input for machine-learning models (Table S17), both the HOMO–LUMO gap and the single-point energy are predicted with markedly reduced accuracy, irrespective of the underlying representation.

Notably, with the exception of SOAP, the predictive performance of RCA spaces for individual HOMO and LUMO energies remains comparatively acceptable, indicating that absolute orbital energies are less sensitive to the geometric distortions induced by equidistance violations than derived properties such as energy gaps.

Table S16: **Mean absolute error for the prediction of the indicated properties using the KRR model using RCA spaces constructed with mono-, bi-, and tricyclic fragments of the N-HPC-1X compounds as the reference constellation.** Green indicates an increase in accuracy compared to the full spaces, yellow indicates a decrease in accuracy of 0.1 eV/a.u. or less, and red indicates a decrease in accuracy of more than 0.1 eV/a.u.

Input	Fragment	HOMO [eV]	LUMO [eV]	GAP [eV]	SPE [a.u.]
CM	Mono	0.233	0.311	0.398	20.576
	Di	0.161	0.173	0.266	9.403
	Tri	NaN	NaN	NaN	NaN
PC3-MAOC	Mono	0.183	0.249	0.342	4.401
	Di	0.128	0.151	0.200	0.148
	Tri	NaN	NaN	NaN	NaN
MAOC	Mono	0.204	0.268	0.384	11.467
	Di	0.176	0.244	0.350	3.221
	Tri	0.143	0.158	0.229	2.022
SOAP	Mono	0.126	0.162	0.217	15.388
	Di	0.067	0.056	0.088	0.526
	Tri	0.064	0.036	0.078	0.159
SLATM	Mono	0.129	0.124	0.166	0.411
	Di	0.067	0.058	0.084	0.056
	Tri	0.045	0.027	0.053	0.041

Table S17: **Mean absolute error for the prediction of the indicated properties using the KRR model using RCA spaces constructed with fragments of the COMPAS-1D compounds as the reference constellation.** Green indicates an increase in accuracy compared to the full spaces, yellow indicates a decrease in accuracy of 0.1 eV/a.u. or less, and red indicates a decrease in accuracy of more than 0.1 eV/a.u.

Input	HOMO [eV]	LUMO [eV]	GAP [eV]	SPE [a.u.]
CM	0.141	0.163	0.303	1790.108
PC3-MAOC	0.138	0.162	0.299	639.634
MAOC	0.133	0.154	0.284	12.697
SOAP	0.128	0.137	0.283	708.050
SLATM	0.089	0.079	0.150	0.173

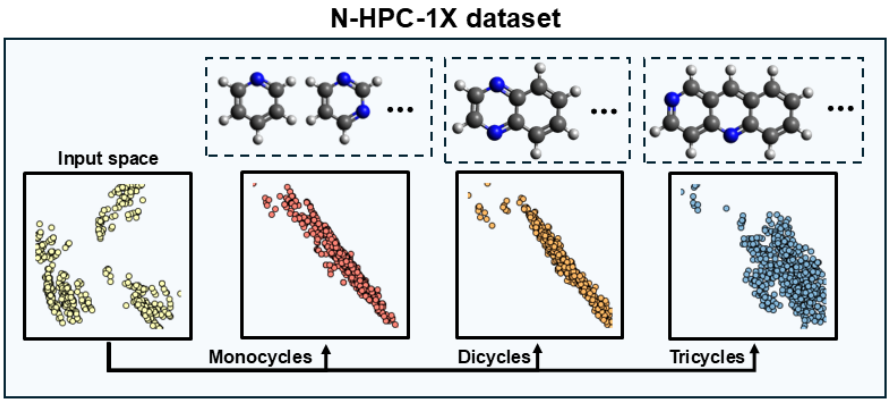

Figure S6: **Schematic visualisation of the transformation in the geometry of chemical space as the composition of the reference constellation is varied.** The SOAP representation was used for the first 500 compounds of the N-HPC-1X dataset.

S8. Impact of the Dimensionality Reduction Technique on the Machine Learning Accuracy of the RCA Space

Reducing the dimensionality of the reference constellation constitutes the only step in RCA that connects the method to conventional dimensionality-reduction techniques. RCA relies on the output of such methods to define a lower-dimensional embedding in which the reference points are positioned, after which multilateration is employed to determine the coordinates of all remaining data points. In the main text, the reduced space of the reference constellation is obtained using PCA. However, the choice of dimensionality-reduction technique is entirely user-defined.

In this section, an alternative strategy is explored in which dimensionality reduction is achieved by assigning each reference point its own axis in the reduced space. In this formulation, the first reference point is represented by the coordinate vector [1, 0, 0, ..., 0], the second by [0, 1, 0, ..., 0], and so forth. In this manner, the applicability of RCA is examined under a limiting scenario in which the reduced space is not derived from any variance-preserving or neighbourhood-preserving method and is therefore entirely decoupled from the structure of the original input space. The predictive accuracy of these RCA spaces for the N-HPC-1X and COMPAS-1D datasets is reported in Tables S18 and S19.

Table S18: **Mean absolute errors of the KRR models obtained using RCA spaces constructed via the unit-vector reduction scheme derived from the specified molecular representations for the N-HPC-1X dataset.** Green indicates an increase in accuracy compared to the full spaces, yellow indicates a decrease in accuracy of 0.1 eV/a.u. or less, and red indicates a decrease in accuracy of more than 0.1 eV/a.u.

Input	HOMO [eV]	LUMO [eV]	GAP [eV]	SPE [a.u.]
CM	0.174	0.160	0.238	11.239
PC3-MAOC	0.144	0.161	0.223	0.064
MAOC	0.154	0.187	0.264	0.059
SOAP	0.151	0.127	0.177	7.094
SLATM	0.101	0.095	0.118	0.053

Table S19: **Mean absolute errors of the KRR models obtained using RCA spaces constructed via the unit-vector reduction scheme derived from the specified molecular representations for the COMPAS-1D dataset.** Green indicates an increase in accuracy compared to the full spaces, yellow indicates a decrease in accuracy of 0.1 eV/a.u. or less, and red indicates a decrease in accuracy of more than 0.1 eV/a.u.

Input	HOMO [eV]	LUMO [eV]	GAP [eV]	SPE [a.u.]
CM	0.100	0.101	0.193	222.171
PC3-MAOC	0.110	0.121	0.228	3.124
MAOC	0.091	0.095	0.180	5.466
SOAP	0.491	0.250	0.467	3648.293
SLATM	0.083	0.086	0.167	0.163

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